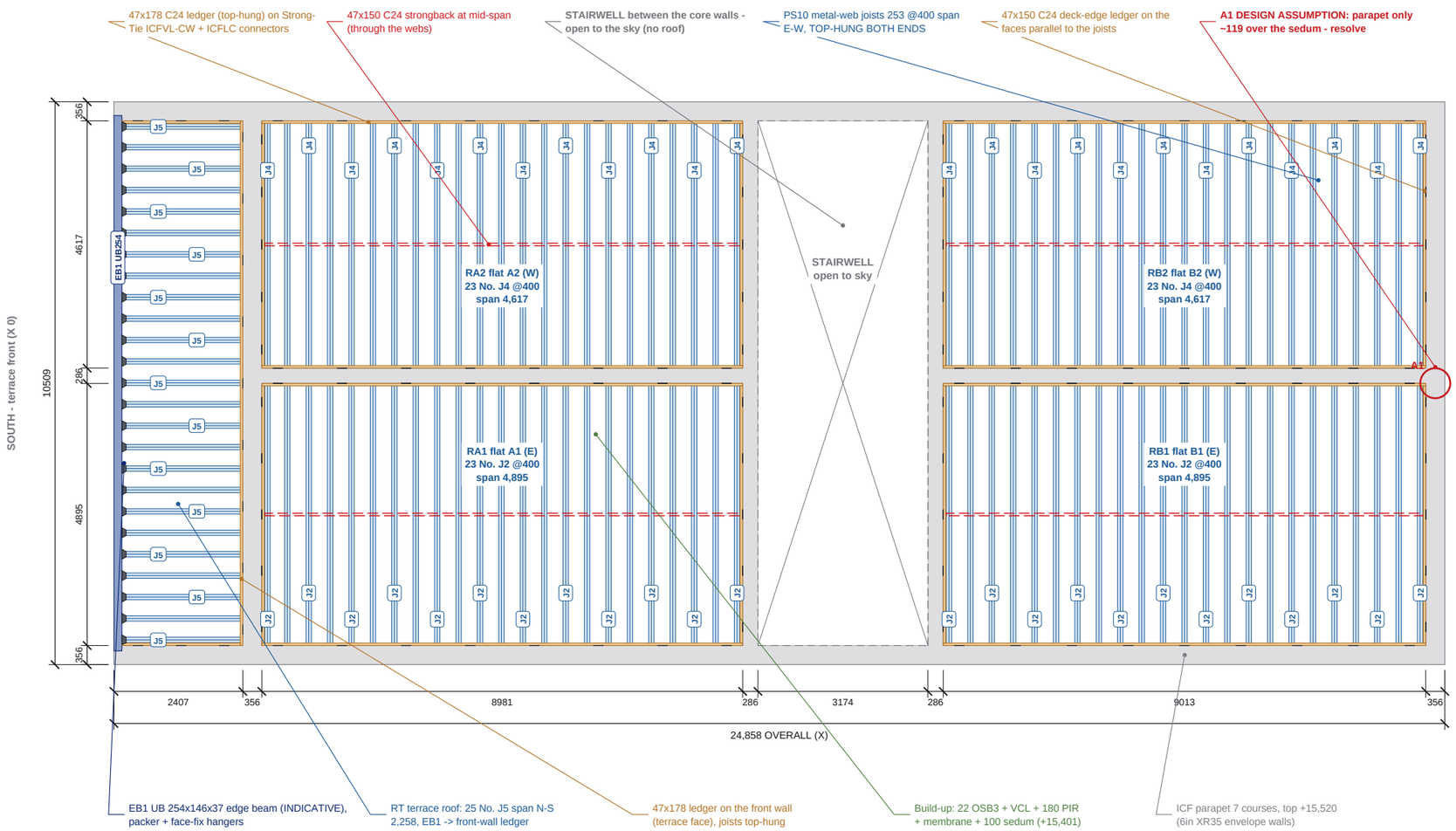
SUPPLY SCOPE (distributor quantities issue, no material prices): Nudura ICF to the new fourth-floor storey only - 6in XR35 external walls + 6in Standard stair-core and separating walls, 7 courses on the steel ring beam - pre-cut by BuildEase, stock forms and accessories by the distributor. The existing building, steelwork, floor, roof, partitions and finishes are shown or assumed in good faith only and must be checked by the relevant professional; the structural engineer to design the ring beam, transfer steels and the walls.

QUOTATION - NOT FOR CONSTRUCTION



FLAT ROOF STRUCTURE PLAN

Scale 1:100 @ A3 | Plan on the joists (deck + build-up omitted), ICF walls cut at +15,300

TRUE NORTH = to the rear (+X)

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

ROOF STRUCTURE SCHEDULES

ROOF JOIST SCHEDULE (253 DEEP, @400)

Type	O/A	Bottom	Bearing	Bays	Qty
J2	4,889	4,795	top-hung both ends	RA1, RB1	46
J4	4,611	4,517	top-hung both ends	RA2, RB2	46
J5	2,252	2,205	hung + top-hung	RT	25
	Total				**117**

O/A = top flange overall; Bottom = bottom flange. Profiles + bearing details on ES26-055-FL-02.

ROOF BUILD-UP

Layer	mm	Top
Joist top (structure)	253	+15,099
22 OSB3 deck	22	+15,121
VCL + 180 PIR (firrings to falls by roofer)	180	+15,301
Single-ply membrane	-	
Sedum / wildflower blanket	100	+15,401
ICF parapet top (7 courses)		+15,520
Parapet upstand over the finish		119 (!)

(!) design assumption to resolve - see note 4.

LEDGERS AND STRONGBACKS (C24)

Item	Length	Qty	Run m
47x150 C24 deck-edge ledger	2,211-4,801	10	41.7
47x150 C24 strongback	8,867-8,897	4	35.5
47x178 C24 ledger (47x225 cut down)	8,981-9,798	9	81.8

METALWORK AND DECK

Item	Qty
Strong-Tie ICFVL-CW + ICFLC ledger connector	123
Galv top-chord restraint clip	209
Face-fix hanger on the EB1 packer	25
22 OSB3 roof deck board (No. / m2)	172 / 193.3

NOTES

1. FLAT SEDUM ROOF, joist top +15,099: 253 deep metal-web joists (PS10 equivalent) @400 c/c spanning ACROSS (E-W) in bays RA1 / RA2 (flats A) and RB1 / RB2 (flats B), TOP-HUNG BOTH ENDS on 47x178 C24 ledgers (47x225 cut down, top +15,052) bolted to the ICF on cast-in Strong-Tie ICFVL-CW + ICFLC connectors (<= 1,200 c/c, 123 No.); galv restraint clips; 47x150 C24 deck-edge ledgers on the faces parallel to the joists; strongbacks at mid-span.
2. COVERED-TERRACE ROOF RT: joists span N-S (model X) from the EB1 UB 254x146x37 edge beam (100 bearing in pockets in the ICF returns; face-fix hangers on a C24 packer in the web) onto the 47x178 ledger on the flats' front wall. EB1 INDICATIVE (370 kg) - structural engineer to design.
3. BUILD-UP: 22 deck + VCL + 180 PIR + membrane + 100 sedum (finish top +15,401, 193.3 m2); firrings to falls, outlets, upstands, edge trims and the green-roof specification by the roofer.
4. DESIGN ASSUMPTION TO RESOLVE: the ICF parapet (7 courses, top +15,520) stands only ~119 above the sedum finish +15,401 - less than the usual 150 minimum membrane upstand and no edge protection. Raise the parapet / upstand, reduce the build-up or detail the capping - architect + roofer to confirm.
5. STAIRWELL between the stair-core walls: open to the sky (no roof structure), as modelled.
6. Joist supplier to design and confirm (saturated sedum, snow, maintenance loads); for pricing only. Levels mm above the rear yard ground (Z 0 ~ +14.92 AOD).

REVISIONS

REV	DATE	DESCRIPTION
-	02.11.26	Quotation issue

Nudura™ BuildEase

CLIENT	PROJECT
Insubloc (distributor) - Tim Evans	2 Castle Gardens, Swansea - new fourth-floor storey (4 flats)

DRAWING TITLE

FLAT ROOF STRUCTURE

Metal-web joists top-hung on ICF ledgers, terrace roof off EB1, sedum build-up

SCALE	SIZE	DATE
1:100	A3	02.11.26
DRAWN	CHECKED	APPROVED
TS		

STATUS	DRAWING No.	REV
QUOTATION	ES26-055-RF-01	-